jmbayes: fast, memory-efficient, data-parallel Bayesian joint models for longitudinal and time-to-event data

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Abstract

We present **jmbayes**, a data-parallel reimplementation of the R package JMbayes2, a widely used tool for Bayesian joint models of longitudinal and time-to-event data. It fits the same model, with the same priors and the same Markov chain Monte Carlo (MCMC) algorithm, and gives empirically equivalent results: the same estimates, within Monte Carlo error. On the largest problem we tested it was 10× faster with 27× less memory, and the gap grew with problem size. For 3,120 patients and 90,000 iterations, JMbayes2 took 104 minutes and 37.8 GB of RAM; jmbayes took 11 minutes and 1.4 GB. jmbayes's memory stayed at 1.3–1.4 GB of RAM and under 0.2 GB of GPU memory in every run, up to 90,000 iterations, while JMbayes2's grew with patients × iterations beyond the capacity of a typical workstation. The gains come from restructuring the computation, not from changing the method: chains and patients are processed as data-parallel tensor operations on a GPU instead of one process per chain, each iteration is replayed as a recorded CUDA graph, and the fit statistics (DIC, WAIC, LPML) are computed exactly from running totals instead of a stored per-patient log-likelihood for every draw. In formal equivalence tests on the primary biliary cholangitis (PBC) data, all 50 model parameters agreed with JMbayes2 within 0.2 posterior standard deviations, and the two programs differed from each other less than two JMbayes2 runs with different random seeds. jmbayes covers JMbayes2's model options (10 marker families, competing risks, multi-state and recurrent events, interval censoring, all functional forms); in 21 further models no parameter was shown to differ, and in 30 repeated runs per program on three data sets every parameter was shown equivalent. jmbayes is open source (GPL-3.0), installable with `pip install jmbayes`, and available with all scripts at `https://github.com/neuratomic/jmbayes`.

1 Introduction

Joint models link repeated measurements of a biomarker to the risk of an event, such as death or transplantation, through shared random effects [3, 2]. They are the standard way to study how a marker's trajectory relates to risk, and they give dynamic predictions that update as new measurements arrive. The R package JMbayes2 [1] fits a broad class of these models in a Bayesian framework, with an MCMC sampler written in C++.

Two practical limits appear as studies grow. First, run time: each MCMC iteration updates every patient's random effects, and models with several markers often need tens of thousands of iterations to converge. Second, memory: JMbayes2 stores the log-likelihood of every patient at every iteration to compute fit statistics after sampling, so its memory grows with patients × iterations. In practice, long runs on cohorts of a few thousand patients no longer fit in the memory of a typical 16 GB workstation.

jmbayes removes both limits without changing the statistical method. It is a port of JMbayes2's sampler, step by step, to PyTorch [4], restructured so that the work of each iteration runs as data-parallel operations. On the same machine, it was 10× faster than JMbayes2 on the largest problems we tested, with a gap that grew with cohort size and run length, and its

memory barely grew with run length. We show with formal equivalence tests that it gives the same results as JMbayes2 within Monte Carlo error.

2 Model

For patient i and marker k, the longitudinal sub-model is a generalised linear mixed model,

$$g_k\{E(y_{ik}(t) \mid b_{ik})\} = \eta_{ik}(t) = x_{ik}(t)^\top \beta_k + z_{ik}(t)^\top b_{ik}, \qquad b_i = (b_{i1}, \ldots, b_{iK}) \sim N(0, D),$$

with an identity link for continuous markers and a logit link for binary ones. The hazard of the event is

$$h_i(t) = h_0(t) \exp\Big\{\gamma^\top w_i + \sum_k \alpha_k \, f_k(\eta_{ik}, t)\Big\},$$

where w_i are baseline covariates and f_k is the functional form that links marker k to the hazard: its current value $\eta_{ik}(t)$, its slope, or its cumulative area. The log baseline hazard is a B-spline with a penalised (random-walk) prior whose precision is estimated. Priors, starting values and all MCMC updates follow JMbayes2: a Metropolis-within-Gibbs sampler with adaptive random-walk proposals for the random effects, fixed effects without a matching random effect, association parameters and spline coefficients, and Gibbs or Metropolis steps for the covariance matrix D and the residual variances. Hierarchical centring of the fixed effects is applied as in JMbayes2.

3 Methodology

Where the parallelism is. We began by classifying the work in one MCMC iteration by the kind of parallelism it allows. The parameter blocks (random effects, covariance matrix, fixed effects, residual variances, baseline hazard, survival and association coefficients) must be updated in order, because each update conditions on the latest values of the others, and iterations must run one after another, because the chain is Markov. That part is sequential. Within each block, however, the same arithmetic is applied to many independent units: every patient's random effects are updated with the same proposal and acceptance rule, every observation's linear predictor and likelihood term has the same form, and the survival integral is evaluated at the same quadrature points for every patient. This is data parallelism: one instruction stream applied to many data elements (SIMD, or SIMT on a GPU). The chains add a second level: they are fully independent, so they are task parallelism (MIMD).

JMbayes2 exploits only the MIMD level: it runs each chain in its own R process on its own CPU core, and each process loops over patients in compiled C++. With the default 3 chains, 3 cores are busy whatever the cohort size, and each process keeps its own copy of the data and of the stored draws. jmbayes recasts the chains as data parallelism as well: it adds the chain as a leading tensor dimension, so a single instruction stream updates all patients of all chains at once. On a GPU, with thousands of parallel lanes, an iteration then costs almost the same for 312 or 3,120 patients until the device is saturated, which is consistent with the 1.4-fold increase in MCMC time we measured for ten times the patients (Section 6). On a CPU, the same code uses vector instructions and threads but far fewer lanes, which is why the CPU version is not faster than JMbayes2. Because iterations themselves cannot be parallelised, the fixed cost of launching many small GPU operations in every iteration matters, which led to the use of CUDA graphs (Section 4).

Reuse instead of new compiled code. JMbayes2 needs its sampler in C++ because loops in R are slow. We asked instead which parts of JMbayes2 already have an equivalent in the Python scientific stack, and wrote only what was missing. Data handling uses pandas; B-spline

bases come from SciPy; the linear mixed models that provide starting values and centre some priors (R's `lme`) are fitted with the REML estimator of statsmodels; starting values for the logistic model come from a statsmodels GLM; numerical optimisation uses SciPy and Gauss–Hermite nodes come from NumPy. Two estimators had no Python version with exactly R's behaviour and were written directly: a binomial mixed model by maximum likelihood with adaptive Gauss–Hermite quadrature (as in R's `GLMMadaptive`) and a Cox model with Efron's handling of ties (as in R's `coxph`). The MCMC sampler itself is written with PyTorch tensor operations.

This removes the need for any compiled code of our own. Python only orchestrates: every numerical operation runs in the compiled back-ends of these libraries, in C++ and CUDA for PyTorch and in C and Fortran (BLAS and LAPACK) for NumPy and SciPy. On the GPU, each MCMC iteration is a single replay of a recorded CUDA graph, so the Python interpreter is not in the inner loop at all. The package is therefore pure Python, distributed as one platform-independent wheel that runs wherever PyTorch runs.

Porting and verification. The sampler was ported from JMbayes2's C++ source function by function, keeping the same update order, proposals, adaptation rules and priors, so that the only intended difference is the random-number stream. Much of this translation was done with the assistance of Claude, a large language model from Anthropic. The author proposed recasting the chains and patients as SIMD data parallelism (above) and directed Claude to profile JMbayes2's code to find where this applies. The author reviewed the code, and correctness was established empirically rather than assumed from the translation: unit tests compare the design matrices, starting fits and priors with JMbayes2's output, and the full model was tested for statistical equivalence with JMbayes2 (Section 5). Where the port revealed apparent inconsistencies in JMbayes2's own code, we reproduced them where possible so that results stay comparable, and documented them (Section 4).

4 Implementation

Batched chains and patients. Every quantity is stored with a leading chain dimension, and all patients' random effects of all chains are updated in one tensor operation. Marker trajectories are updated incrementally when random effects change instead of being rebuilt.

CUDA graphs. An MCMC iteration launches many small GPU kernels. jmbayes records one full iteration as a CUDA graph the first time it runs and replays the recorded graph afterwards, which avoids most of the per-kernel launch overhead. All state stays on the GPU; only the kept parameter draws are copied back.

Fit statistics without the log-likelihood table. Let $\ell_i^{(s)}$ be the log-likelihood of patient i (markers, event and random effects) at kept draw s, for $s = 1, \ldots, S$. JMbayes2 stores the $S \times n$ matrix of $\ell_i^{(s)}$ and computes, after sampling,

$$\begin{aligned}
\mathrm{DIC} &= \bar{D} + p_D, \quad \bar{D} = -\tfrac{2}{S}\textstyle\sum_s \sum_i \ell_i^{(s)}, \quad p_D = \bar{D} - D(\bar{\theta}), \\
\mathrm{WAIC} &= -2\textstyle\sum_i \log\left(\tfrac{1}{S}\sum_s e^{\ell_i^{(s)}}\right) + 2\sum_i \mathrm{Var}_s\left(\ell_i^{(s)}\right), \\
\mathrm{LPML} &= \textstyle\sum_i \log \mathrm{CPO}_i, \quad \mathrm{CPO}_i = \left(\tfrac{1}{S}\sum_s e^{-\ell_i^{(s)}}\right)^{-1}
\end{aligned}$$

[5, 6, 7]. Each of these needs only a few running totals per patient: $\sum_s \ell_i^{(s)}$; $\log \sum_s e^{\ell_i^{(s)}}$ and $\log \sum_s e^{-\ell_i^{(s)}}$, accumulated on the log scale with a stable log-add-exp (the exponentials themselves under- or overflow); and the variance of $\ell_i^{(s)}$, accumulated with Welford's algorithm [8]

and combined across chains with the pairwise formula of Chan et al. [9]. jmbayes updates these totals at every kept draw and never stores the matrix, so memory depends on the number of patients but not on the run length. The totals are kept in 64-bit precision; the values agree with those computed from the full matrix to within 10^{-8} relative error. For analyses that need every draw, such as PSIS-LOO and its Pareto-k diagnostic [10], the matrix can still be kept with `fit(..., save_loglik=True)`.

Precision. jmbayes samples in 32-bit floating point by default and refits automatically in 64-bit if a run produces invalid values. Over 9 matched benchmark settings on the GPU, 32-bit was 1.67× faster than 64-bit (geometric mean; 95% CI 1.49–1.86), and its results were equivalent to JMbayes2 in the same tests as 64-bit (Section 5). We rejected bfloat16: it keeps only 8 significant bits, so near a log-likelihood of 1,000 neighbouring representable values are 4 apart, too coarse for MCMC acceptance decisions.

Possible issues in JMbayes2. Porting code line by line, and testing every result against it, is a close reading, and it turned up places where JMbayes2 (0.6-5) appears not to do what it intends. These are our reading of the code and of the outputs we observed; we may have misunderstood the intent in some cases, and we are reporting them to the maintainers. Where results stay comparable, jmbayes reproduces the behaviour so that the comparisons test the port and not the fix.

- *Sampler.* A random-effects density does not appear to be refreshed after the fixed-effects update, nor a survival denominator after the smoothing-parameter update. jmbayes reproduces both.
- *Dynamic prediction.* When sampling a new patient's random effects, the survival term appears to be scaled differently from the code that then computes the risk. jmbayes uses the consistent version by default and reproduces JMbayes2's with `predict(..., r_sampling=True)`; for PBC patient 93 the one-year risk was 0.63 instead of 0.70.
- *Hierarchical centring with splines.* With a spline of time in a fixed-effect interaction (e.g. `ns(year, 2) * sex` with random `~ ns(year, 2)`), the centring multiplier appears to be computed from the spline evaluated at 1 rather than from the random-effect column, so the fixed-effects step no longer leaves the linear predictors unchanged. In one test a single fixed-effect draw changed a patient's association term 300-fold. jmbayes corrects this (models without such terms are unaffected, bit for bit), and the comparisons use additive spline models.
- *Smaller points.* `Dexpit()` appears to be read as `expit()` (transformation names are matched case-sensitively); `ms_setup()` with a single covariate appears to place values on other patients' rows; and the separate mixed-model fits that `jm()` starts from and centres some priors on did not always converge (Student's t, censored normal, beta-binomial), so these comparisons start both programs from R's separate fits. For binomial markers with several trials and beta-binomial markers, JMbayes2's fit statistics looked implausible (conditional p_D about half of the DIC; marginal DIC 0, LPML ∞, WAIC missing); we have not traced the cause.

Details and the code locations are in the repository's `NOTES.md`.

Interface. Models are written with Python objects (`Marker`, `Event`, `JointModel`) or with JMbayes2-style formula strings; both build the same model and give identical draws. jmbayes does not impute or silently drop data: missing or invalid values stop the fit with a message naming the columns.

Table 1: Hazard ratios from the same model and data: posterior mean and 95% interval (mean ± 1.96 posterior SD of the log hazard ratio), two seeds averaged per program, as in every figure and table of this paper. Markers act through their current value.

Effect on the hazard of death or transplantation	JMbayes2	jmbayes
Female sex	0.52 (0.26–1.03)	0.51 (0.26–1.03)
log serum bilirubin (per unit)	1.68 (1.20–2.36)	1.68 (1.20–2.35)
Prothrombin time (per second)	1.01 (0.80–1.27)	1.01 (0.79–1.28)
Ascites (per unit log-odds)	1.77 (1.35–2.31)	1.77 (1.36–2.31)

5 Equivalence with JMbayes2

Setup. We used the PBC data [11] (312 patients, 1,945 visits, 169 deaths or transplants) and the model from the JMbayes2 documentation: a survival model with sex, linked through the current value of log serum bilirubin and prothrombin time (linear mixed models) and ascites (logistic mixed model), each with a random intercept and slope. Each program was run twice with different seeds, with 12 chains of 30,000 iterations (5,000 burn-in), giving 300,000 kept draws per run.

Error bars. To compare MCMC estimates we need their Monte Carlo standard errors (MCSE). Comparing two runs of the same program, the MCSE reported by JMbayes2 understated the real run-to-run error: only 80% (R) and 88% (Python) of differences fell within ±2 MCSE. We therefore inflated all MCSEs by the factor $k = 1.40$ estimated from these same-program pairs; the correction applies equally to both programs.

Test. For each of the 50 parameters, d is the difference between the jmbayes and JMbayes2 posterior means (each pooled over its two runs). The programs are declared equivalent for that parameter if the 90% confidence interval of d lies within $\pm\delta$ (two one-sided tests at $\alpha = 0.05$ [12]), with δ equal to 0.1 or 0.2 posterior standard deviations (SD).

Results. All 50 parameters were equivalent at ±0.2 SD and 43 at ±0.1 SD; none was shown to differ. The median absolute difference between jmbayes and JMbayes2 was 0.011 SD (largest 0.042), smaller than between two JMbayes2 runs (0.023 SD, largest 0.141) and two jmbayes runs (0.018 SD, largest 0.070). All 15 regression coefficients led to the same conclusions in both programs (direction, significance at 5%, and whether the 95% credible interval excludes zero). Table 1 and Figure 1a show the hazard ratios.

Fit statistics agreed within run-to-run variation (relative differences below 0.3%). Convergence was also the same: the Gelman–Rubin $\hat{R}$ [13] of every regression coefficient was below 1.05 in all four runs, the largest being 1.043 in both programs, for the same parameters (those of the ascites sub-model). With JMbayes2's default length (3 chains, 3,500 iterations), neither program converged for the ascites parameters, at any cohort size.

For dynamic predictions, we gave jmbayes JMbayes2's own posterior draws; with JMbayes2's random-effects sampling (`r_sampling=True`), jmbayes's predicted risks matched JMbayes2's to within 0.01. With its default, consistent sampling, risks can differ by a few percentage points; for PBC patient 93, the one-year risk was 0.63 instead of 0.70.

Competing risks. jmbayes also fits cause-specific hazards for competing events, the model JMbayes2 fits with `crisk_setup()` and `strata()`: each cause has its own baseline hazard, covariate effects and marker associations, and predictions give the cumulative incidence of each cause. We fitted death and transplantation in PBC as competing events, with cause-specific

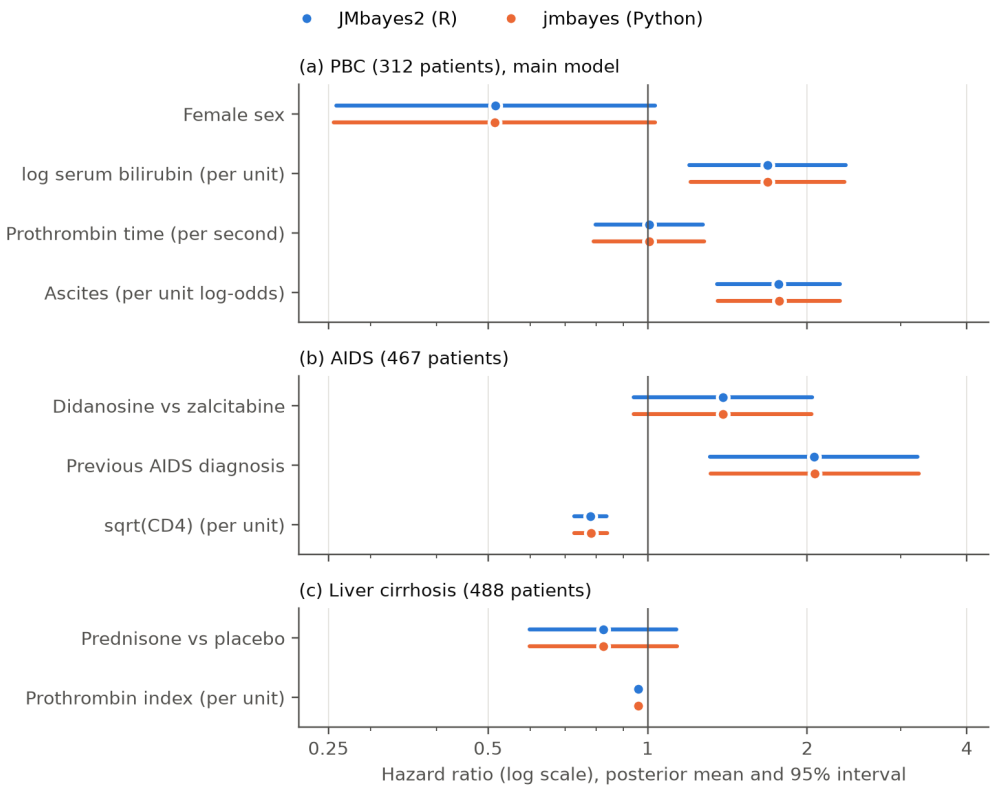


Figure 1: Hazard ratios in JMbayes2 and jmbayes on the three data sets: (a) the main PBC model of this section, (b, c) the two other data sets that ship with JMbayes2 (Section 5). Posterior mean and 95% interval (mean ± 1.96 posterior SD of the log hazard ratio), two seeds averaged per program.

effects of sex and of the current values of log serum bilirubin and prothrombin time, in both programs (two seeds each, 12 chains of 30,000 iterations). None of the 50 parameters was shown to differ; 36 were shown equivalent within ±0.2 posterior SD and 27 within ±0.1 SD, the rest being inconclusive at this run length. The largest difference between the programs, 0.065 SD, was smaller than between two JMbayes2 runs with different seeds (0.18 SD). Hazard ratios agreed to two decimals; for example, per unit of log bilirubin, 3.27 (JMbayes2) and 3.27 (jmbayes) for death and 3.66 and 3.64 for transplantation. JMbayes2 0.6-5's `predict()` fails on this model, so cumulative incidences were checked internally: summed over causes, they matched one minus the overall survival to within 0.004.

Prediction accuracy. jmbayes also ports JMbayes2's measures of predictive accuracy for dynamic predictions: time-dependent ROC curves and AUC, the Brier score (also integrated over the prediction window), and calibration (ICI, E50, E90 and the calibration curve), with model-based or inverse-probability-of-censoring weights. Given the same predicted risks, they reproduced JMbayes2's values to 10^{-10} or better in two prediction windows.

Other markers, strata and late entry. jmbayes 0.6 adds Poisson, negative binomial and Student's t markers, stratified baseline hazards and delayed entry (left truncation), with JMbayes2's densities, priors and quadrature (a 7-point Gauss–Kronrod rule on $[\text{entry}, T_i]$ under delayed entry). We tested two models as above (two seeds, 12 chains of 30,000 iterations). In the first, the baseline hazard was stratified by treatment, entry was set to 20% of follow-up, and log serum bilirubin and prothrombin time were Gaussian markers; all 47 parameters were

shown equivalent within ±0.1 SD (largest difference 0.032 SD). In the second, platelets were negative binomial, rounded serum bilirubin Poisson and prothrombin time Student's t with 4 degrees of freedom; all 50 parameters were shown equivalent within ±0.1 SD (largest 0.034 SD). Because GLMMadaptive's Student's t fit did not converge and `jm()` centres some priors on the separate fits, both programs started from R's separate fits in the second model. MCMC took 1,712 s and 3,907 s in JMbayes2 and 135 s and 173 s in jmbayes.

Further features. jmbayes 0.7 and 0.8 add the rest of JMbayes2's model options: beta, gamma, censored normal, beta-binomial and binomial (with trials) markers; time-varying survival covariates (counting-process rows); interval censoring; recurrent events with a terminal event and a frailty; competing risks combined with delayed entry and strata; effects per stratum; all functional forms (velocity, acceleration, area and Delta with time windows, transformations such as expit and Dexp, interactions with covariates, and time-varying associations); Weibull, piecewise-constant and natural-spline baseline hazards; multi-state models; case weights; ridge and horseshoe shrinkage priors; and marginal fit statistics. We tested each with one model, in both programs, with two seeds of 8 chains and 15,000 iterations, using the same equivalence test. To check data other than PBC, we also fitted a model to each of two other data sets that ship with JMbayes2: CD4 counts and death in 467 HIV patients (`aids`), and the prothrombin index and death in 488 patients with liver cirrhosis (`prothro`).

Table 3 gives the results, together with the main model and the competing-risks model above, so that every model in this paper is judged the same way. No parameter in any of the 21 further models was shown to differ between the programs. Per parameter, the equivalence tests (Table 5 in the appendix) showed all or nearly all parameters equivalent within ±0.2 SD in 16 models; in the other five (competing risks with delayed entry and strata, recurrent events, effects per stratum, the beta-binomial model and the multi-state model) many were inconclusive, but two runs of the same program differed about as much as the two programs did, which points to slow mixing at this run length rather than to a difference. In the beta-binomial model, for example, the random intercepts of the two markers correlate at 0.98, so their associations with the hazard are nearly confounded.

Figure 2a compares all 89 hazard ratios of all 23 models in this paper (each listed in Table 6 in the appendix). MCMC estimates carry Monte Carlo error, so even two JMbayes2 runs with different seeds do not give identical hazard ratios; the yardstick is therefore whether jmbayes differs from JMbayes2 more than JMbayes2 differs from itself. It does not (Figure 2b): between single runs, 70% of hazard ratios agreed within 1% for JMbayes2 against itself with another seed, and 74% for JMbayes2 against jmbayes (median difference 0.39% and 0.26%). The two distributions coincide, so the differences are the Monte Carlo error that any rerun of JMbayes2 shows. The larger differences are in the slowly mixing models, where the two programs' own runs also differ, most of all in the beta-binomial model, whose associations are nearly confounded.

Other data sets. The tests above, like the main comparison, use the PBC data. To check that agreement does not depend on that data set, we fitted a joint model to each of the two other data sets that ship with JMbayes2. In the AIDS data (467 patients randomised to didanosine or zalcitabine), the square root of the CD4 count, with a random intercept and slope over time, was linked to death, adjusting for drug and a previous AIDS diagnosis. In the liver cirrhosis data (488 patients randomised to prednisone or placebo), the prothrombin index was linked to death, adjusting for treatment. In both, all 22 parameters were shown equivalent within ±0.2 SD (18 and 19 within ±0.1 SD), and the hazard ratios agreed to the third decimal (Figure 1b and c, drawn and computed exactly as for PBC in panel a); for example, 1.39 in both programs for didanosine and 0.83 in both for prednisone.

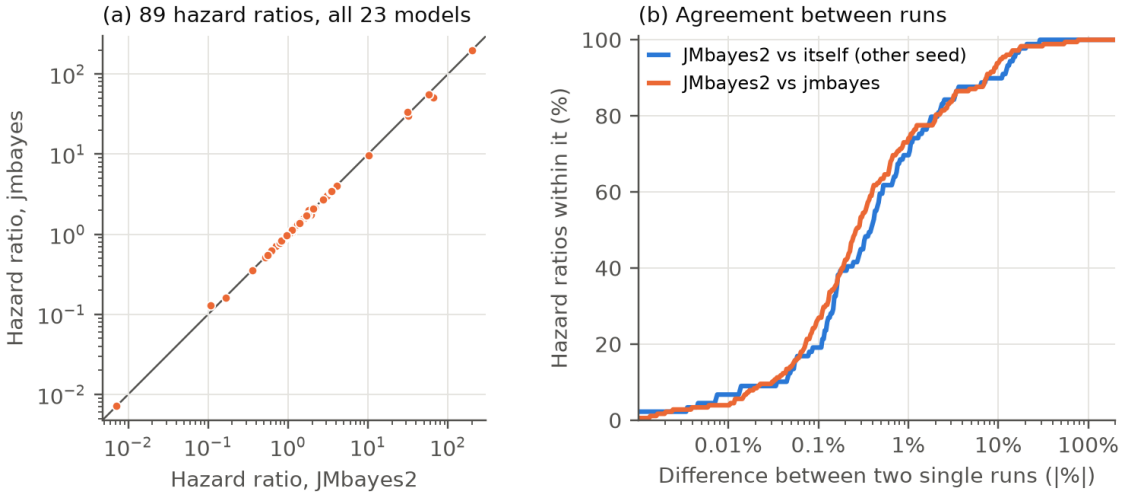


Figure 2: (a) All hazard ratios of all 23 models, jmbayes against JMbayes2 (posterior means, two seeds averaged; the line is equality). (b) Agreement between single runs: the share of hazard ratios (y) whose relative difference is at most x, for JMbayes2 against itself run with another seed (blue) and JMbayes2 against jmbayes (orange). The curves coincide: jmbayes differs from JMbayes2 no more than JMbayes2 differs from itself.

Repeated runs. Two runs per program can show a parameter equivalent only where the chains mix quickly; elsewhere the result stays inconclusive. To test equivalence with known statistical power, we ran each program 30 times on each of the three data sets, with different seeds (the main PBC model and the AIDS and cirrhosis models above, 8 chains of 15,000 iterations each). The runs are independent, so the spread of their posterior means is the Monte Carlo error itself, and two-sample tests on the 30 + 30 runs need no assumptions about autocorrelation within chains. For each parameter we tested whether the two programs' posterior means and posterior SDs differ (Welch's t-test, and a Kolmogorov–Smirnov test of the whole distribution of the run means), with Benjamini–Hochberg correction across parameters, and whether they are equivalent (two one-sided tests).

All 94 parameters were shown equivalent, in their posterior mean within ± 0.1 posterior SD and in their posterior SD within $\pm 10\%$; 87 were also shown equivalent within the stricter ± 0.05 SD (Table 2). No parameter was shown to differ after correction. Before correction, 9 of the 94 means had $p < 0.05$, where about 5 are expected by chance; with 30 runs per program the tests detect very small differences, and the largest was 0.048 posterior SD, less than the run-to-run SD of a single run of either program. That spread was the same in both programs (median 0.052 and 0.047 posterior SD for PBC, 0.020 and 0.021 for AIDS, 0.015 and 0.013 for cirrhosis, JMbayes2 first), so jmbayes mixes as well as JMbayes2. From the observed spread, 30 runs per program gave at least 80% power to show equivalence within ± 0.1 SD for every parameter (the most demanding needed 22).

The comparisons also found an error in jmbayes: for markers whose density is a difference of large log-gamma terms (beta-binomial, negative binomial, beta and gamma), 32-bit rounding exceeded the signal at large dispersions, and the beta-binomial dispersion drifted upwards. These densities are now computed in 64-bit precision, and the affected comparisons were rerun. On the AIDS data, statsmodels' default optimiser stopped at a poor optimum of the separate linear mixed model (its REML log-likelihood 739 units below the maximum), which moved the priors centred on that fit; jmbayes now also tries Powell's method and keeps the better fit, and the results in Table 3 use this. These runs shared the machine with each other, so we do not report their timings; Section 6 measures speed.

Table 2: Repeated-runs comparison: each program run 30 times per data set with different seeds (8 chains of 15,000 iterations). Per parameter, the 30 posterior means and SDs from jmbayes are compared with the 30 from JMbayes2. Shown equivalent: two one-sided Welch tests at 5%, for the posterior mean within ±0.1 and ±0.05 posterior SD, and for the posterior SD within ±10%. Shown to differ: Welch's t-test (mean, SD) and the two-sample Kolmogorov–Smirnov test (whole run distribution of the posterior mean), after Benjamini–Hochberg correction at 5% false discovery rate. 80% power: parameters for which 30 runs per program give at least 80% power to show equivalence of the mean within ±0.1 SD.

		Shown equivalent			Shown to differ			
		Mean		SD			Run	80%
Data set	Param.	±0.1	±0.05	±10%	Mean	SD	distr.	power
PBC (312 patients, 3 markers)	50	50	43	50	0	0	0	50
AIDS (467 patients)	22	22	22	22	0	0	0	22
Liver cirrhosis (488 patients)	22	22	22	22	0	0	0	22

Table 3: jmbayes against JMbayes2 for every model in this paper, judged the same way: two runs per program with different seeds (12 chains of 30,000 iterations for the first four models, 8 of 15,000 for the rest). Shown to differ: parameters whose difference is shown to exceed 0.2 posterior SD (two one-sided tests at 5%). Difference: between a JMbayes2 run and a jmbayes run, per parameter, in units of its posterior SD (median and largest over the parameters). Equivalence counts per model are in Table 5; Section 5 repeats the comparison with 30 runs per program.

		Shown to	Difference (posterior SDs)	
Model	Parameters	differ	Median	Largest
Main model (PBC, 3 markers)	50	0	0.020	0.096
Competing risks (PBC)	50	0	0.021	0.068
Strata + late entry	47	0	0.015	0.033
NB, Poisson, Student's t	50	0	0.012	0.037
Time-varying covariate	23	0	0.021	0.037
Interval censoring	22	0	0.024	0.080
Competing risks + entry + strata	60	0	0.111	0.231
Effects per stratum	50	0	0.044	0.180
Recurrent events	38	0	0.103	0.154
Gamma, beta, cens. normal	50	0	0.019	0.063
Binomial (trials), beta-binomial	25	0	0.130	0.500
Velocity, area	27	0	0.045	0.115
expit, Dexp, acceleration	27	0	0.055	0.120
Value by sex, Delta	27	0	0.034	0.071
Time-varying association	26	0	0.022	0.089
Weibull baseline	12	0	0.010	0.122
Piecewise-constant baseline	20	0	0.014	0.041
Natural-spline baseline	21	0	0.011	0.038
Multi-state	18	0	0.028	0.140
Case weights	22	0	0.014	0.057
Shrinkage priors	39	0	0.019	0.103

Model	Parameters	Shown to differ	Difference (posterior SDs)	
			Median	Largest
AIDS data	22	0	0.027	0.068
Prothrombin data	22	0	0.021	0.079

6 Speed and memory

Setup. To vary the cohort size, the PBC data were copied 1, 2, 5 and 10 times with new patient identifiers (312 to 3,120 patients). We fitted the same model with 3 chains (burn-in 500, thinning 5) at 3,500 iterations (JMbayes2's default) and, for 1,560 and 3,120 patients, up to 90,000 iterations. Each run was a separate process, run one at a time. Peak RAM is the resident memory of the whole process tree, sampled every 0.2 s, which includes JMbayes2's worker processes (one per chain). MCMC time excludes the initial separate model fits, which both programs need. JMbayes2 runs its chains in parallel on the CPU; jmbayes ran on the GPU in 32-bit precision. All runs used one machine: Intel Core i9-12900K (x86-64; 16 cores, 8 performance and 8 efficiency, 24 threads), 64 GB RAM, NVIDIA GeForce RTX 3090 (24 GB), Windows 11, R 4.6.2 with JMbayes2 0.6-5, Python 3.11 with PyTorch 2.6 (CUDA 12.4).

Results. Table 4 and Figure 3 show the results. jmbayes was 3.3× faster on 312 patients and 8.0–8.6× faster on 1,560–3,120 patients at the default length, and 9.7–10.5× on long runs: at 90,000 iterations for 3,120 patients, JMbayes2 took 104 minutes and 37.8 GB of RAM, and jmbayes 11 minutes and 1.4 GB. Ten times more patients increased jmbayes's MCMC time 1.4-fold and JMbayes2's 3.7-fold. jmbayes's RAM stayed at 1.3–1.4 GB and its GPU memory below 0.2 GB in every run, up to 90,000 iterations, while JMbayes2's memory grew with both patients and iterations.

The speed comes from the GPU. Run on the CPU alone, jmbayes was about as fast as JMbayes2 on small cohorts and slower on large ones (445 s vs 202 s for 3,120 patients at 3,500 iterations), but used less than half the memory (0.6–0.7 GB).

7 Limitations

Each benchmark setting was run once, so single timings carry noise; for example, JMbayes2 took less time for 3,120 patients at 14,000 iterations than for 1,560, and its run at 60,000 iterations overlapped with other work (Table 4). Larger cohorts were made by copying one data set, which tests scaling but not new data. Equivalence was tested in depth on one model and data set; three further models (splines, slope and area associations, fixed effects without random effects) agreed in smaller tests, and each later feature (marker families, strata, delayed entry, competing risks, interval censoring, recurrent events, functional forms, baseline hazards, multi-state models, case weights and shrinkage priors) was tested on one or two models, mostly of the same data. The speed-up needs a GPU that PyTorch drives through its CUDA interface: NVIDIA (tested here) or AMD with ROCm (not tested). Apple and Intel GPUs are not supported yet; there jmbayes runs on the CPU. On a CPU it is about as fast as JMbayes2, because it runs all chains in one process, which uses only the data parallelism. Combining both levels of Section 3, with groups of chains in separate processes (MIMD, as JMbayes2) and patients batched within each (SIMD), should make the CPU version faster too; we leave this to future work. jmbayes covers JMbayes2's model options, with three exceptions: survival covariates that are functions of the time variable (an interaction in the association gives the same time-varying effects), interval censoring or recurrent events combined with delayed entry or competing risks, and dynamic

Table 4: MCMC time and peak RAM. jmbayes on the GPU in 32-bit precision; JMbayes2 on the CPU. Dashes: not run.

		JMbayes2		jmbayes		
Patients	Iterations	Time	RAM	Time	RAM	Speed-up
312	3,500	55 s	1.28 GB	17 s	1.28 GB	3.3×
624	3,500	87 s	1.45 GB	17 s	1.28 GB	5.0×
1,560	3,500	157 s	1.95 GB	20 s	1.29 GB	8.0×
3,120	3,500	202 s	2.41 GB	24 s	1.31 GB	8.6×
1,560	7,000	237 s	2.34 GB	41 s	1.29 GB	5.8×
1,560	14,000	716 s	3.36 GB	82 s	1.30 GB	8.8×
1,560	28,000	1,046 s	6.25 GB	164 s	1.31 GB	6.4×
3,120	7,000	450 s	3.44 GB	48 s	1.31 GB	9.4×
3,120	14,000	521 s	6.15 GB	96 s	1.32 GB	5.4×
3,120	28,000	2,056 s	12.0 GB	196 s	1.33 GB	10.5×
3,120	60,000	5,893 s[a]	25.2 GB	429 s	1.39 GB	
1,560	90,000	–	–	529 s	1.36 GB	
3,120	90,000	6,266 s	37.8 GB	643 s	1.38 GB	9.7×

[a] This run overlapped with other work on the machine. JMbayes2's time per iteration was 98 ms, against 70–73 ms at 28,000 and 90,000 iterations, so this time is likely inflated and no speed-up is given.

predictions for recurrent-event models. These stop with an error rather than fitting a different model.

8 Availability

jmbayes 0.8.0 is available from PyPI (`pip install jmbayes`) and at `https://github.com/neuratomic/jmbayes` under the GPL-3.0 licence, like JMbayes2. The repository contains the scripts and results for every number in this paper, including the full equivalence report. jmbayes is an independent reimplementation and is not affiliated with or endorsed by the authors of JMbayes2, whose work it builds on.

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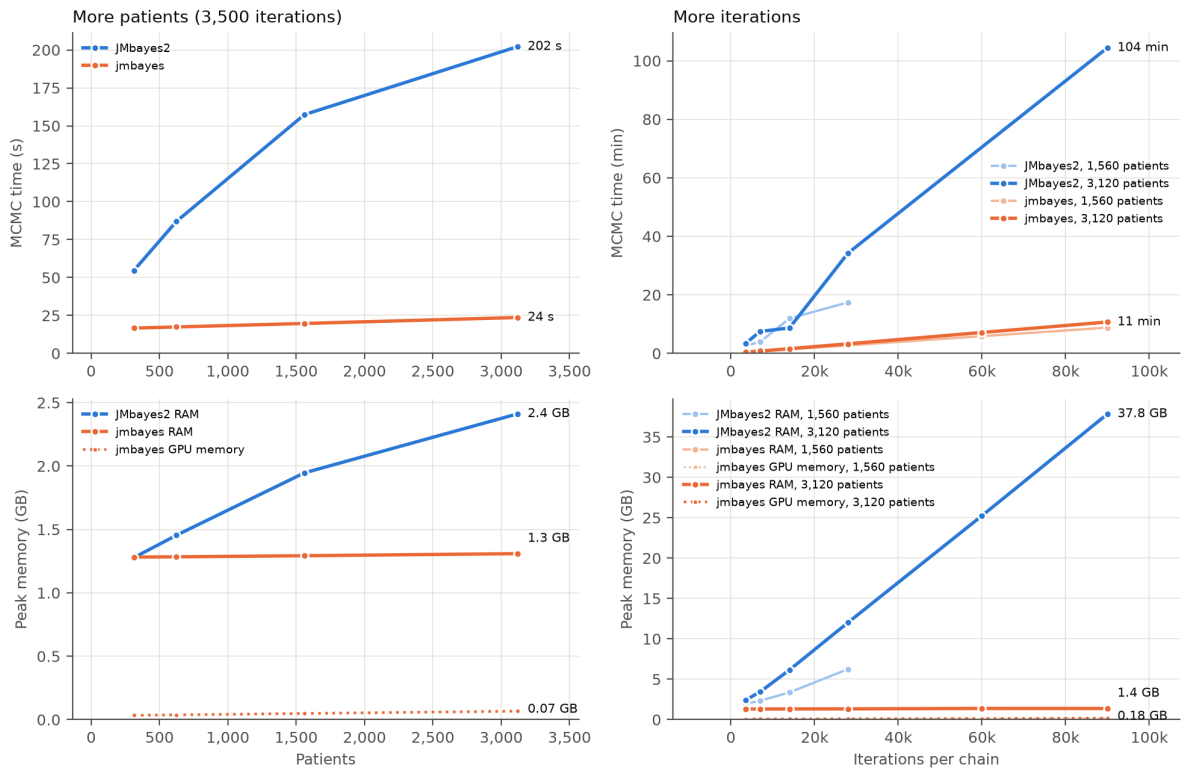


Figure 3: MCMC time (top) and peak memory (bottom) as the number of patients (left, 3,500 iterations) and the number of iterations (right; light lines 1,560 patients, dark lines 3,120) grow. Dotted: jmbayes GPU memory.

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A Equivalence tests per model

Table 5: Equivalence tests per model (the runs of Table 3). Each parameter is either shown equivalent (within ±0.2 posterior SD; the ±0.1 count is a subset), shown to differ (by more than 0.2 SD), or inconclusive: two runs per program are too few to separate a difference below 0.2 SD from Monte Carlo error, which happens where the chains mix slowly. No parameter was shown to differ. Largest difference: between the programs (R vs Python, seeds averaged) and between the two JMbayes2 runs of the same model (R vs R).

		Equivalent		Incon-		Largest diff.	
Model	Param.	±0.1	±0.2	clusive	Differ	R–Py	R–R
Main model (PBC, 3 markers)	50	43	50	0	0	0.042	0.141
Competing risks (PBC)	50	27	36	14	0	0.065	0.181
Strata + late entry	47	47	47	0	0	0.032	0.086
NB, Poisson, Student's t	50	50	50	0	0	0.034	0.062
Time-varying covariate	23	23	23	0	0	0.026	0.051
Interval censoring	22	17	22	0	0	0.077	0.113
Competing risks + entry + strata	60	12	21	39	0	0.231	0.269
Effects per stratum	50	26	34	16	0	0.056	0.121
Recurrent events	38	12	23	15	0	0.154	0.107
Gamma, beta, cens. normal	50	46	50	0	0	0.038	0.108
Binomial (trials), beta-binomial	25	6	7	18	0	0.244	0.139
Velocity, area	27	9	25	2	0	0.115	0.219
expit, Dexp, acceleration	27	20	25	2	0	0.120	0.060
Value by sex, Delta	27	22	27	0	0	0.071	0.041
Time-varying association	26	25	26	0	0	0.045	0.087
Weibull baseline	12	10	12	0	0	0.025	0.114
Piecewise-constant baseline	20	20	20	0	0	0.031	0.046
Natural-spline baseline	21	21	21	0	0	0.023	0.056
Multi-state	18	12	12	6	0	0.115	0.316
Case weights	22	22	22	0	0	0.057	0.039
Shrinkage priors	39	26	37	2	0	0.045	0.273
AIDS data	22	18	22	0	0	0.064	0.032
Prothrombin data	22	19	22	0	0	0.041	0.129

B Hazard ratios of every model

Table 6: Hazard ratios (posterior means, two seeds averaged) in JMbayes2 and jmbayes; up to 4 per model (all are in the repository).

Model	Term	JMbayes2	jmbayes
Main model (PBC, 3 markers)	`sexfemale`	0.517	0.515
	`value(log(serBilir))`	1.683	1.681
	`value(prothrombin)`	1.008	1.006
	`value(ascites)`	1.767	1.769
Competing risks (PBC)	`sexfemale [transplanted]`	2.169	2.140
	`sexfemale [dead]`	0.702	0.703
	`value(log(serBilir)) [transplan...`	3.658	3.644
	`value(log(serBilir)) [dead]`	3.271	3.269

Model	Term	JMbayes2	jmbayes
Strata + late entry	sexfemale	0.811	0.809
	value(log(serBilir))	2.972	2.965
	value(prothrombin)	1.203	1.203
NB, Poisson, Student's t	sexfemale	0.909	0.907
	value(platelets)	0.580	0.584
	value(bili_count)	3.607	3.603
	value(prothrombin)	1.444	1.448
Time-varying covariate	sexfemale	0.675	0.676
	edema_tv	3.248	3.242
	value(log(serBilir))	2.892	2.895
Interval censoring	sexfemale	0.896	0.896
	value(log(serBilir))	3.521	3.528
Competing risks + entry + strata	sexfemale [transplanted]	2.071	1.999
	sexfemale [dead]	0.750	0.755
	value(log(serBilir)) [transplan...	3.546	3.464
	value(log(serBilir)) [dead]	3.100	3.098
Effects per stratum	sexfemale [placebo]	0.777	0.781
	sexfemale [D-penicil]	0.865	0.876
	value(log(serBilir)) [placebo]	3.083	3.092
	value(log(serBilir)) [D-penicil]	3.407	3.408
Recurrent events	group [R]	1.615	1.616
	group [T1]	1.571	1.566
	age [R]	1.052	1.052
	age [T1]	1.049	1.048
Gamma, beta, cens. normal	sexfemale	0.705	0.707
	value(serBilir)	3.058	3.052
	value(alb_p)	0.007	0.007
	value(lalk)	0.622	0.623
Binomial (trials), beta-binomial	sexfemale	1.105	1.059
	value(count2)	0.107	0.128
	value(count3)	65.843	50.888
Velocity, area	sexfemale	0.836	0.835
	value(log(serBilir))	1.801	1.985
	velocity(log(serBilir), eps = 0...	32.075	30.115
	area(log(serBilir), time_windo...	1.920	1.744
expit, Dexp, acceleration	sexfemale	0.779	0.774
	vexpit(value(log(serBilir)))	198.969	195.992
	acceleration(log(serBilir))	31.266	33.295
	vexp(value(log(serBilir))):slop...	1.330	1.331
Value by sex, Delta	sexfemale	0.549	0.544
	Delta(log(serBilir), time_wind...	10.120	9.674
	value(log(serBilir)):sexmale	2.735	2.738
	value(log(serBilir)):sexfemale	3.389	3.404
Time-varying association	sexfemale	0.862	0.863
	value(log(serBilir))	4.020	4.022
	value(log(serBilir)):ns(year, k...	0.799	0.800
	value(log(serBilir)):ns(year, k...	0.824	0.827
Weibull baseline	sexfemale	0.831	0.832
	value(log(serBilir))	3.423	3.419
Piecewise-constant baseline	sexfemale	0.852	0.852

Model	Term	JMbayes2	jmbayes
	value(log(serBilir))	3.485	3.484
Natural-spline baseline	sexfemale	0.880	0.873
	value(log(serBilir))	3.392	3.387
Multi-state	X [1]	1.563	1.563
	X [2]	0.784	0.784
	X [3]	1.770	1.770
	value(y) [1]	1.704	1.704
Case weights	sexfemale	0.853	0.854
	value(log(serBilir))	3.484	3.485
Shrinkage priors	sexfemale	1.041	1.038
	age	1.039	1.039
	drugD-penicil	0.954	0.950
	value(log(serBilir))	1.423	1.452
AIDS data	drugddI	1.386	1.386
	prevOIAIDS	2.060	2.065
	value(CD4)	0.780	0.781
Prothrombin data	treatprednisone	0.825	0.825
	value(pro)	0.959	0.959